Determination of Winning Baseball Strategies Under MLB's Extra-Inning Rule Modifications

Baseball Track
ID: 124

1. Introduction

As part of Major League Baseball's 2020 health and safety protocols instituted in response to the COVID-19 pandemic, three rule changes were made before the start of the shortened season.[1] While the universal designated hitter and position players pitching rules certainly have left their impact on professional baseball, perhaps the most thought-provoking change from a strategy standpoint was a new rule placing a runner on second to begin extra innings.[2] The new rule saw each half-inning after the conclusion of the ninth begin with an automatic runner on second base, with the runner being the player in the batting order immediately preceding that half-inning's leadoff hitter, or a pinch-runner.[3]

Coined the "designated runner," and later the "automatic runner," this rule remained in a temporary fashion for the 2021 and 2022 seasons.[4] Since the change was identified as one that "significantly affects terms and conditions of Players' employment," and therefore classified as a part of the "Playing Rules," the MLB's Joint Competition Committee voted in favor of permanent amendment before the start of the 2023 season.[5]

Prior to its implementation at the Major League level, this rule had been a part of Minor League Baseball (MiLB) dating back to the 2018 season.[6] In fact, the concept was first tested during the 2017 World Baseball Classic, which instituted an "automatic runner" beginning in the 11th inning; it was later tested in the Rookie Level Gulf Coast and Arizona Leagues in that same year.[7] Despite just two years of practice, considerable results were produced from extra-innings games in the 2018 and 2019 Minor League seasons. In those two seasons, 73% of MiLB extra-inning games ended in the 10th inning, whereas in the two seasons prior (2016-2017), only 45% of MiLB extra-inning games ended in the 10th inning.[8] These results represent the crux of the argument for the implementation of the rule in the MLB. By placing a runner on second base at the start of each

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[1] Major League Baseball, "MLB Rule Changes for 2020," *MLB.com*, accessed 2025, https://www.mlb.com/news/mlb-rule-changes-for-2020.
[2] Major League Baseball, "MLB Rule Changes for 2020."
[3] "Designated Runner," *MLB.com Glossary*, accessed 2025, https://www.mlb.com/glossary/rules/designated-runner.
[4] "Designated Runner," MLB.com Glossary.
[5] Major League Baseball Players Association, *2022–2026 Basic Agreement* (2022), https://www.mlbplayers.com/_files/ugd/4d23dc_d6dfc2344d2042de973e37de62484da5.pdf.
[6] Bill Walker, "New for 2020: MLB Extra Inning Rule," *MiLB.com* (August 30, 2020), https://www.milb.com/news/major-league-baseball-extra-inning-rule.
[7] Walker, "New for 2020: MLB Extra Inning Rule."
[8] Walker, "New for 2020: MLB Extra Inning Rule."

inning after the ninth, games would be shortened, pitchers would be more protected due to fewer pitches being thrown, and more excitement would be added to the game.

While the reasoning behind the implementation of the rule change is seemingly positive, there have been mixed reactions from players. Those expressing positivity appreciate the reduced strain on pitchers' arms and the chance to end games sooner, arguing that the game's pace is more appealing.[9] On the other side, detracted from the challenge and strategy that is integral to baseball, arguing the essence of the game is rooted in its unpredictability, and starting with a runner on second may feel gimmicky.[10] However, while the new extra-innings situation removes some traditional aspects of baseball strategy, it also introduces an entirely new set of strategic considerations that managers can employ to give their team an advantage when necessary. This paper examines different strategies managers should consider during the most common extra innings situations and seeks to determine which are most effective.

2. Run Expectancy

Driving the strategies of a manager from an analytical standpoint is the modern concept of a run expectancy matrix. This will be key to successful analysis and interpretation in this paper, as the matrix plays a pivotal role in understanding winning baseball strategies, even beyond the extra-innings framework. The run expectancy matrix can be derived from any sample of game-level data. By recording the base-out state of each play and how many runs scored from that point until the end of the inning, run expectancy averages can be developed. The base-out state for any given play consists of two components: the runners on base and the number of outs. Runners on base are often displayed in the following format: 0-0-0 (no runners on base), 1-0-0 (runner on first base), 0-2-3 (runners on second and third base), and so on. The number of outs is added to build a complete base-out state (e.g., 0 (outs)-1-0-0, 2-0-2-0, 1-1-0-3).

21-24 Matrix			
Runners On	0 Outs	1 Out	2 Outs
---	0.5	0.27	0.1
1--	0.9	0.54	0.23
-2-	1.14	0.71	0.33
--3	1.37	0.98	0.38
12-	1.51	0.94	0.46
1-3	1.82	1.19	0.51
-23	2.04	1.41	0.57
123	2.38	1.63	0.82

Also known as RE24 to reflect the twenty-four base-out states possible in a game, encompassing eight base states and three out states, the matrix quantifies the number of runs a team is expected to score: "a mathematical description of how likely teams is to score in a given situation, in aggregate."[11] Run expectancy matrices are fundamentally based on Markov chains, which model "a

[9] Zachary Miller, "Does a Runner Start on 2nd Base in Extra Innings? Understanding MLB's Extra-Inning Rules," *Plate Crate Blog*, accessed 2025, http://platecrate.com/blogs/baseball-101/does-a-runner-start-on-2nd-base-in-extra-innings-understanding-mlbs-extra-inning-rules.
[10] Miller, "Does a Runner Start on 2nd Base?"
[11] Dan Zimmerman, "The Run Expectancy Matrix Reloaded for the 2020s," *FanGraphs*, accessed 2025, https://blogs.fangraphs.com/the-run-expectancy-matrix-reloaded-for-the-2020s/.

sequence of events where the probability for each outcome only depends on the previous state of the system."[12]

Assigning a value to certain scenarios in a game allows for a concrete understanding of the value any result may have. Not only can this shed light on what an event means for a run environment, but it also clarifies whether a strategy may be favorable or not. Thus, while an extra-innings state may offer the same 24 possible base-out states, the nature of a half-inning starting with an automatic runner on second calls for a revisitation of what run expectancy may mean.

Often, plays in the ninth inning and extra innings are not recorded to be included in a matrix, as again, this is when run scoring is most likely to be biased toward a certain target number of runs. Based on our knowledge of run expectancy and dissimilar late-game strategy, our hypotheses before analysis include that the run expectancy matrix will possibly yield similar results when visiting teams are batting but yield divergent results when home teams are batting. Furthermore, we predict that in many scenarios, traditional small-ball strategies will yield less favorable outcomes resulting in a lower run expectancy.

3. Common Extra Innings Strategy Considerations

There are many considerations when understanding winning baseball strategies under MLB's extra-inning rule modifications. Never in Major League Baseball had there been a fixed scenario with a runner on base to start new half-innings, opening the door for new methods of run-scoring tactics. A baseline understanding of different strategies that could be employed is necessary for familiarity with the forces of run expectancy in this new environment.

Perhaps the most notable strategic change with the new rule is the vastly different settings of the visiting and home teams. While the visiting team may have an advantage in preparation due to encountering the same situation of a runner on second, no outs, and a tie game, the same level of doubt that comes with not knowing how many runs need to be scored still applies. Meanwhile, the home team is aware of exactly the number of runs they need to generate but must be slightly more prepared to know what to do, as they may face a tie game or a game in which they trail.

3.1. Attempting a Sacrifice Bunt

One strategy a team may choose to employ is to have the first batter of the half-inning attempt a sacrifice bunt. The most likely outcome of this choice, should the ball be put in play, is the trade of an out for the baserunner advancing from second to third. At first glance, this trade-off may appear to yield a lower run expectancy overall. However, this strategy can be beneficial in certain situations, especially for a home team that is looking to score only one run in their half-inning. After all, the runner on third will be ninety feet from scoring, meaning that a hit, error, sacrifice fly, wild pitch, or even an infield ground ball will allow for the team to reach their goal.

[12] Calestini, Lucas. "The Elegance of Markov Chains in Baseball." *Sports Analytics* (Medium). September 10, 2018. https://medium.com/sports-analytics/the-elegance-of-markov-chains-in-baseball-f0e8e02e7ac4.

3.2. Automatic Runner Attempting to Steal Third Base

Another way a team can look to gain an advantage is through aggressive baserunning and having the automatic runner attempt to steal third base. If successful, the advantage to the offensive team is clear. With a runner on third base and less than two outs, the ways to score that runner greatly increase compared to when the runner was on second base. Not only are the chances of scoring the runner with a base hit virtually guaranteed, but possible outcomes that result in the batter recording an out can also score a run from third base. This strategy may be more useful after a team has already recorded a leadoff out, as they would be more desperate to advance a runner to third before two outs have been made.

3.3. Intentionally Walking the First Batter of the Half-Inning

In addition to strategy shifts for the batting team, the changed extra innings rule introduces possible changes to the strategies of the pitching team. More specifically, how to pitch to the first batter of the half-inning with a runner already on second base. One strategy that is implemented in many scenarios is intentionally walking the first batter to set up the situation of zero outs with runners on first and second base. While this puts an additional runner on base, it also increases the likelihood of inducing a double-play with an increased number of force outs available.

4. Data and Methodology

To derive winning strategies under the recently appended MLB extra innings rules, this paper will seek to develop an extra innings-specific run expectancy matrix using data from extra innings games played during the 2021 to 2024 regular season. Since, as previously mentioned, the automatic runner rule is not implemented during postseason play; these games will not be included in the analysis. Furthermore, these four seasons were selected to allow for comparison to Ben Clemens' Run Expectancy matrix, which averages the individual season run expectancy matrix for 2021 through 2024 to create an aggregate matrix.

To conduct this analysis, game-level data were obtained from Retrosheet, a publicly available baseball historical database founded in 1989 for the purpose of computerizing play-by-play accounts, box scores, and event files for all American League and National League games dating back to 1910. Retrosheet data consists of play-by-play event files that track game sequences by home and visiting teams, which include pitch sequence, batter plate appearance results such as walk, out, hit, etc. The robust data has roster information, box scores, and game logs. Using this data, it is possible to reconstruct base/out states, run expectancy, and win probabilities. Using an in-house Python-based Retrosheet game scraper, analysis consists of querying all games within a series of games and locating games that resulted in extra innings, finding base-out states, and computing run expectancies. These run expectations in matrix form are computed for the top and bottom of the inning.

5. Created Run Expectancy Matrices and Observations

The introduction of the automatic runner produced measurable changes to the structure and outcomes of extra innings games, altering what makes up a winning strategy in this new environment. Because run-scoring dynamics shift under these conditions, examining how run expectancy changes is crucial for developing winning strategies. This section presents run expectancy matrices generated using Retrosheet play-by-play data. While the focus of this study is extra innings, matrices for each half of the ninth inning are also included because of their typical exclusion from traditional run expectancy matrices, as well as providing a point of comparison under similar conditions. Furthermore, separate matrices for the top and bottom halves of each inning are shown because of the unique run scoring conditions present in each. To provide context on the differences in run-scoring conditions, comparisons will be made between the generated matrices and Ben Clemens' aggregated matrix for the 2021-2024 seasons.

5.1. Inning: 9 – Side: Top

The unique run-scoring conditions of the top of the ninth inning are a result of two factors: the objective of the batting team and the lack of an automatic runner. In each preceding inning, the main objective for the batting team will always be to maximize their total run output. While this is still true in the ninth inning, unlike in extra innings, the automatic runner is not implemented. Therefore, the run-scoring environment in the top of the ninth inning can be treated similarly to that of the aggregated run expectancy for all innings played before the ninth.

Top 9 Matrix			
Runners On	0 Outs	1 Out	2 Outs
---	0.3942	0.2033	0.0771
1--	0.7245	0.3969	0.1625
-2-	0.9801	0.52	0.2387
--3	1.4889	0.9185	0.3517
12-	1.2792	0.8337	0.4011
1-3	1.5	0.9965	0.4486
-23	1.7679	1.2787	0.5978
123	2.2797	1.5068	0.7686

T9 vs. 21-24			
Runners On	0 Outs	1 Out	2 Outs
---	-0.1058	-0.0667	-0.0229
1--	-0.1755	-0.1431	-0.0675
-2-	-0.1599	-0.19	-0.0913
--3	0.1189	-0.0615	-0.0283
12-	-0.2308	-0.1063	-0.0589
1-3	-0.32	-0.1935	-0.0614
-23	-0.2721	-0.1313	0.0278
123	-0.1003	-0.1232	-0.0514

However, when comparing these two matrices, it is observed that run expectancy is much lower in the top of the ninth inning than in each preceding inning. This can be attributed to the opposing pitchers used at this point in the game complicating a team's ability to score. If the home team is ahead, the visiting team is likely to see one of the best opposing relief pitchers, if not the opposing closer, on the mound. Having to face these high-leverage relievers results in a lower run expectancy in the ninth inning than in each inning preceding it, when the batting team may face the opposing team's starting pitcher or a lower-quality relief pitcher. Even if the home team is losing or the game is tied, their goal of allowing the fewest possible runs will result in a higher-quality relief pitcher being used–that is, unless the run differential is so wide that the home team's manager opts to leave his better relievers in the bullpen.

Only two base-out states have greater run expectancy values in the top of the ninth inning than the aggregated matrix (no outs with a runner on third base, two outs with runners on second and third base), and the absolute value of both differences are less than the average absolute value of all other base-out states (0.1255). This further proves that fewer runs are scored on average in the top of the ninth inning than in any other inning, despite the opportunity to score as many runs as possible.

5.2. Inning: 9 – Side: Bottom

The unique run-scoring conditions of the bottom of the ninth inning, should it be played, present a different run scoring environment from every other half-inning played for the same reasons as discussed for the top of the ninth inning—the lack of an automatic runner and the quality of relief pitchers likely faced. In fact, the likelihood of a high-leverage reliever being used is even higher than in the top of the ninth. If the bottom of the ninth is played, this means the game is either tied or the visiting team is winning at the start of the inning. Therefore, the visiting team's manager will be more inclined to utilize their best relievers to extend the game and send it to extra innings or win the game. Like the top of the ninth inning, this environment is proven by the comparison between the run expectancies of the bottom ninth inning and the aggregated matrix. None of the base-out states in the bottom of the ninth inning have a higher run expectancy than its counterpart in the aggregated matrix.

Bottom 9 Matrix			
Runners On	0 Outs	1 Out	2 Outs
---	0.3338	0.1821	0.0767
1--	0.5541	0.3321	0.1395
-2-	0.789	0.5069	0.2428
--3	1.1053	0.8	0.3725
12-	0.9646	0.563	0.3476
1-3	1.4211	0.8517	0.3309
-23	1.3605	1.0047	0.5177
123	1.56	1.0941	0.562

B9 vs. 21-24			
Runners On	0 Outs	1 Out	2 Outs
---	-0.1662	-0.0879	-0.0233
1--	-0.3459	-0.2079	-0.0905
-2-	-0.351	-0.2031	-0.0872
--3	-0.2647	-0.18	-0.0075
12-	-0.5454	-0.377	-0.1124
1-3	-0.3989	-0.3383	-0.1791
-23	-0.6795	-0.4053	-0.0523
123	-0.82	-0.5359	-0.258

B9 vs. T9			
Runners On	0 Outs	1 Out	2 Outs
---	-0.0604	-0.0212	-0.0004
1--	-0.1704	-0.0648	-0.023
-2-	-0.1911	-0.0131	0.0041
--3	-0.3836	-0.1185	0.0208
12-	-0.3146	-0.2707	-0.0535
1-3	-0.0789	-0.1448	-0.1177
-23	-0.4074	-0.274	-0.0801
123	-0.7197	-0.4127	-0.2066

While like each preceding half inning where the automatic runner is not in effect, the bottom of the ninth inning is unique as the batting team does not have the opportunity to maximize their total run output, instead only being able to score enough runs to tie or win the game. This can be seen through a comparison of run expectancy in each half of the ninth inning. Only two base-out states have higher run expectancy values in the bottom of the ninth inning than the top of the ninth inning matrix (two outs with a runner on second base, two outs with a runner on third base), and the absolute value of both differences is less than the average absolute value of all other base-out states (0.1876). This further proves the assertion that fewer runs are scored on average in the bottom half of the ninth than the top due to of the inability to maximize run output.

5.3. Inning: 10 – Side: Top

Top 10 Matrix			
Runners On	0 Outs	1 Out	2 Outs
---	0.4319	0.2362	0.1038
1--	0.6485	0.3042	0.1418
-2-	0.9195	0.4362	0.3114
--3	2	0.6944	0.3088
12-	1.5283	0.7333	0.4231
1-3	1.5758	0.8667	0.4353
-23	0.9333	0.8824	0.4545
123	2.1852	1.8571	0.7576

The unique run-scoring conditions of the top of the 10th can be explained by the implementation of the automatic runner in combination with the batting team having the opportunity to score as many runs as possible. Therefore, the top of the 10th inning has a unique run expectancy.

When examining the matrix for the top of the 10th inning, it is important to note that the state of zero runners on base with zero outs can largely be ignored. Since the implementation of the automatic runner, this state is incredibly rare in extra innings. The only way for this to be achieved would be for the automatic runner to record an out or score prior to the first plate appearance of the inning being completed.

While this can happen through a combination of the runner being picked off, stolen base attempts (both successful and unsuccessful), wild pitches/passed balls, or balks, it is very unlikely.

T10 vs. 21-24			
Runners On	0 Outs	1 Out	2 Outs
---	-0.0681	-0.0338	0.0038
1--	-0.2515	-0.2358	-0.0882
-2-	-0.2205	-0.2738	-0.0186
--3	0.63	-0.2856	-0.0712
12-	0.0183	-0.2067	-0.0369
1-3	-0.2442	-0.3233	-0.0747
-23	-1.1067	-0.5276	-0.1155
123	-0.1948	0.2271	-0.0624

Finally, the implementation of the amended extra inning rules comes into play. To examine the automatic runner's effect on run expectancy, differences in the comparison of run expectancy between the top of the 10th inning and the aggregated matrix, and the comparison of the top of the ninth inning and the aggregated matrix, are observed.

Like the expectancies for both halves of the ninth inning, the run expectancy for each state in the top of the 10th inning is lower when compared to the 2021-2024 average expectancy matrix, this time in all but four states (zero outs with a runner on third, zero outs with runners on first and second, one out with the bases loaded, and two outs with no runners on base). However, this is double the number of states with a greater run expectancy than seen in the comparison of the top of the ninth inning and the aggregated matrix. This increase can be attributed to the existence of the automatic runner.

T10 vs. T9			
Runners On	0 Outs	1 Out	2 Outs
---	0.0377	0.0329	0.0267
1--	-0.076	-0.0927	-0.0207
-2-	-0.0606	-0.0838	0.0727
--3	0.5111	-0.2241	-0.0429
12-	0.2491	-0.1004	0.022
1-3	0.0758	-0.1298	-0.0133
-23	-0.8346	-0.3963	-0.1433
123	-0.0945	0.3503	-0.011

Two of the base-out states where run expectancy is higher during the top of the 10th inning than the aggregated matrix are particularly notable. The first of these states, zero outs with a runner on third, is also higher in the top of the ninth inning than in the aggregated matrix, yet the run expectancy in the top of the 10th inning is 0.5111 runs greater, a 429.86% increase, than it is in the ninth inning. The second state, zero outs with runners on first and second, is different from its counterpart in the ninth inning has a lower run expectancy than the aggregated matrix. The expectancy in the top of the 10th inning is 0.2491, which runs higher than in the top of the ninth inning, a 19.47% increase. The "T10 vs. T9" run expectancy matrix shows a complete comparison of the run expectancies between the two innings.

5.4. Inning: 10 – Side: Bottom

Bottom 10 Matrix			
Runners On	0 Outs	1 Out	2 Outs
---	0.227	0.1179	0.0667
1--	0.3535	0.2357	0.1484
-2-	0.7174	0.2754	0.1308
--3	1	0.6667	0.3478
12-	0.7805	0.4578	0.2674
1-3	0.9231	0.7436	0.1395
-23	0.6	1.2963	0.72
123	0.5	0.4737	0.4444

The unique run-scoring conditions of the bottom of the 10th inning can be explained by the implementation of the automatic runner in combination with the batting team not having the opportunity to score as many runs as possible. Therefore, the bottom of the 10th inning has a unique run expectancy.

Like the top half of the 10th inning, the state of zero runners on base with zero outs can largely be ignored due to the unlikelihood of it occurring. This is especially true if the home team is losing by more than one run at the start of the 10th inning. If losing by multiple runs, the home side will not be as aggressive attempting to score the automatic runner because they need more than that run to reach a target of tying the game, let alone winning the game. This greatly decreases the chances the automatic runner will attempt to steal third base and, as a result, also decreases the chance of him being picked off.

B10 vs. T10			
Runners On	0 Outs	1 Out	2 Outs
---	-0.2049	-0.1183	-0.0371
1--	-0.295	-0.0685	0.0066
-2-	-0.2021	-0.1608	-0.1806
--3	-1	-0.0277	0.039
12-	-0.7478	-0.2755	-0.1557
1-3	-0.6527	-0.1231	-0.2958
-23	-0.3333	0.4139	0.2655
123	-1.6852	-1.3834	-0.3132

In addition, like the relationship between the two halves of the ninth inning, the majority of the states in the bottom of the 10th have a lower expectancy than those in the top of the 10th due to the scoring environment, meaning the home team does not have the opportunity to score as many runs as possible, only the amount that will either tie or win the game. Just four states in the bottom of the 10th have a higher run expectancy than their counterparts in the top of the 10th (one out with runners on second and third, two outs with a runner on first, two outs with a runner on third, two outs with runners on second and third). Furthermore, when comparing the run expectancy in the bottom of the 10th to the average run expectancy from 2021 to 2024, just one state (two outs with runners on second and third) has a higher expectancy.

B10 vs. 21-24			
Runners On	0 Outs	1 Out	2 Outs
---	-0.273	-0.1521	-0.0333
1--	-0.5465	-0.3043	-0.0816
-2-	-0.4226	-0.4346	-0.1992
--3	-0.37	-0.3133	-0.0322
12-	-0.7295	-0.4822	-0.1926
1-3	-0.8969	-0.4464	-0.3705
-23	-1.44	-0.1137	0.15
123	-1.88	-1.1563	-0.3756

6. Strategy Evaluation

In this section, the validity of the most common strategies employed by teams in extra innings will be explored in the context of the calculated run expectancies. The strategy section of this paper discussed three major considerations. From the offensive point of view, bunting to lead off the inning or aggressively stealing third base are points of emphasis. From the standpoint of the defense, intentionally walking the leadoff batter yields the trade-off of another baserunner for the potential of a ground-ball double play.

6.1. A Note About the Ninth Inning

As previously mentioned, the objective for the batting team in the top of the ninth inning is to maximize their total run output. The strategies used by a team to score should not change; however, teams need to take advantage of run-scoring opportunities because runs are more difficult to come by at this stage of the game when high-leverage relievers are in the game. The comparison between the top of the ninth inning run expectancy and the aggregated run expectancy from 2021 to 2024 shows scoring runs to be much more difficult in the ninth inning.

6.2. Attempting a Sacrifice Bunt in the First Plate Appearance

Here, the effects on run expectancy in the top and bottom half of the 10th inning of having the first batter of the half-inning attempt a sacrifice bunt will be explored.

6.2.1. Top of the 10th Inning

For a visiting team in the top of the 10th, the first strategy to consider is having the first batter of the inning attempt a sacrifice bunt with the hope of moving the runner from second to third base. However, using the matrix, bunting in the first at bat of the top of the 10th inning is not a winning strategy by examining the relationship between the run expectancy at the start of the inning (0.9195), and the possible states resulting from a sacrifice bunt attempt. If the sacrifice is not successful and the automatic runner remains at second base, assuming the batter records an out, run expectancy falls by 0.4833, a 52.56% decrease. Another possibility would be an unsuccessful sacrifice attempt resulting in the automatic runner being tagged out. Assuming the batter safely makes it to first base, run expectancy falls by 0.6153, a 66.92% decrease. Even when the sacrifice bunt is successful, meaning the batter records an out while the automatic runner makes it to third base safely, run expectancy still falls by 0.2251, a decrease of 24.48%. Only in the rare scenario of a bunt attempt resulting in neither the automatic runner nor the batter recording an out does the run expectancy increase. While this increase of 0.6563, or 71.38%, seems large, it does not offset the total risk taken by bunting. The most likely outcome of having the first batter attempt a sacrifice bunt is that it is successful, moving the runner to third. Even in this outcome, the run expectancy decreases, meaning it is not a winning strategy.

6.2.2. Bottom of the 10th Inning

For a home team in the bottom of the 10th, this same strategy of having the first batter of the inning attempt a sacrifice bunt, with the hope of moving the runner from second to third base, can be analyzed. From the matrix, bunting in the first at bat of the top of the 10th inning is not a winning strategy by examining the relationship between the run expectancy at the start of the inning (0.7174), and the possible states resulting from a sacrifice bunt attempt. If the sacrifice is not successful and the automatic runner remains at second base, assuming the batter records an out, run expectancy falls by 0.4420, a 61.61% decrease. Another possibility would be an unsuccessful sacrifice attempt resulting in the automatic runner being tagged out. Assuming the batter safely makes it to first base, run expectancy falls by 0.4817, a 67.14% decrease. Even when the sacrifice bunt is successful, meaning the batter records an out while the automatic runner makes it to third base safely, run expectancy still falls by 0.0507, a decrease of 7.07%. Only in the rare scenario of a bunt attempt resulting in neither the automatic runner nor the batter recording an out does the run expectancy increase, which does not offset the total risk taken by bunting. The most likely outcome of having the first batter attempt a sacrifice bunt is that it is successful, moving the runner to third. Even in this outcome, the run expectancy decreases, meaning it is not a winning strategy for the home team either.

6.3. Having the Automatic Runner Attempt to Steal Third Base

Here, the effect on run expectancy in the top and bottom half of the 10th inning of having the automatic runner attempt to steal third base will be explored.

6.3.1. Top of the 10th Inning

In the top half of the 10th inning, it is rare to see the automatic runner attempt to steal third base with zero outs because of the goal of maximizing total run output. Narratively, it does not make sense to risk a runner that is already in scoring position running into an out just to move him up from second to third base. However, the results of the run expectancy matrix for this half inning suggest it could be a prudent move. Should the automatic runner successfully steal third base, run expectancy increases from 0.9195 to 2.000, a 117.51% increase. Conversely, if the automatic runner is caught stealing, run expectancy would decrease from 0.9195 to 0.2362, a 74.31% decrease.

Therefore, in a scenario with a very fast runner on second base, the potential gain of scoring a run could outweigh the risk of being thrown out at third. The strategy a manager should employ is dependent on the personnel on the field.

6.3.2. Bottom of the 10th Inning

In the bottom half of the 10th inning, it is possibly more likely to see the automatic runner attempt a steal of third base. This is largely dependent on the game situation, as the home team is not looking to maximize their total run output but reach a target. If the score is tied or the home team is trailing by only one run, it could make sense to take the risk of advancing a runner that is already in scoring position. However, the results of the run expectancy matrix for this half inning suggest taking this risk is not worth it. Should the automatic runner successfully steal third base, run expectancy increases from 0.7147 to 1, a 39.92% increase. Conversely, if the automatic runner is caught stealing, run expectancy would decrease from 0.7147 to 0.1179, an 83.50% decrease. Even in a scenario with a very fast runner on second base, the potential gain of scoring a run is far lower than the decrease in run expectancy experienced if the stolen base attempt is unsuccessful. Should the home team be losing by multiple runs, the recommendation to not attempt a steal of third base would be even stronger.

6.4. Intentionally Walking the First Batter of the Inning

Here, the effects on run expectancy in the top and bottom half of the 10th inning of intentionally walking the first batter of the half-inning will be explored.

6.4.1. Top of the 10th Inning

From the home team's perspective, consideration could be given to intentionally walking the first batter of the inning to increase the likelihood of a double-play. In the context of the Top of the 10th Run Expectancy matrix, this would result in an increase of the visitor's run expectancy by 0.6088, a 66.21% increase. This would suggest that intentionally walking the batter is detrimental to the home team's goal of preventing a run from scoring. However, if a manager believes their chances of getting the first batter of the inning out while not advancing the runner are very slim, while also liking their chances of inducing the next batter to hit into a double play, they could elect to intentionally walk the batter as the resulting run expectancies show a benefit to the home side. Using the matrix, this can be expressed as the run expectancy with 2 outs and a runner on third base (0.3088) is lower than the expectancy with one out and a runner on second base (0.4362).

6.4.2. Bottom of the 10th Inning

From the visiting team's perspective, consideration could also be given to intentionally walking the first batter of the inning to increase the likelihood of a double-play. In the context of the Bottom of the 10th Run Expectancy matrix, this would result in an increase in the home team's run expectancy by only 0.0631, an 8.08% increase. This would suggest that intentionally walking the first batter of the inning is not necessarily detrimental to the visiting team's goal of preventing a certain number of runs, especially if the visiting team manages to score multiple runs in the top half of the 10th inning. Regardless of who the leadoff batter is, this finding presents a notable strategy consideration for managers to think about.

7. Conclusion

The implementation of MLB's automatic-runner rule has undoubtedly reshaped extra innings strategic decision-making to win a game. Using run expectancy matrices generated from Retrosheet play-by-play data from regular-season games in the 2021 through 2024 seasons, this paper provides half-inning-specific matrices of how run scoring changes in extra innings, and how teams can take advantage of the unique conditions presented by each half-inning.

A consistent pattern of run expectancy in extra innings being lower than in preceding innings emerged, largely because the automatic runner narrows run-scoring outcomes and increases the intensity of every at-bat. However, the consequences of strategic decisions made for the most common situations differ between the visiting and home teams. Visiting teams, free to score as many runs as possible, benefit from strategies that increase the potential to do so. Conversely, home teams are only able to score enough runs needed to tie or win the game and therefore face narrower but more volatile strategic considerations.

Outside of rare cases, attempting a sacrifice bunt in the first plate appearance of either half of an extra inning, despite the perceived advantage of moving the automatic runner to third, consistently decreases run expectancy for both the visiting and home teams. Similarly, because of the risks involved, having the automatic runner attempt to steal third base can only be justified under certain personnel conditions. This is particularly true for home teams that only need to score one run. For the pitching team, intentionally walking the first batter of the inning can provide a benefit, depending on that team's ability to induce a double play. These findings dispute more traditional thinking but emphasize the importance of probabilistic decision-making in extra innings.

This analysis demonstrates that the amended extra innings rule does more than shorten games and reduce stress on pitchers; it reshapes the strategic logic of extra innings decision-making. Teams that maximize the value of outs, avoid unnecessary risk, and adjust their approach based on inning context rather than traditional small-ball conventions will be rewarded. As more games are played with the automatic runner in place, run expectancy will remain a critical tool for determining winning strategies.

7.1. Relationship Between Run Expectancy and Win Probability

When using the extra innings run expectancy matrices, the limitation of attempting to define the relationship between run expectancy and win probability exists. In other words, translating a run expectancy value to an exact increase in win probability. Whether a team is in a tie game, winning, or losing, it is an undisputable fact that as a team scores more runs, their probability of winning that game increases. This is a result of the independence of run expectancy. However, the win probability of one team in extra innings can be dependent on that team's opponent. For a visiting team, this relationship remains independent because of their objective to maximize the total number of runs scored. With the game being tied at the beginning of each new inning after the ninth, there is no "target" a visiting team needs to reach. However, for the home team, the increase in win probability of each action taken depends on the number of runs scored in the top half of the inning. While one action may increase the home team's run expectancy, it may not increase their win probability.

7.2. Personnel Considerations

While the extra innings run expectancy matrices provide a good starting point for determining a winning strategy, it does not consider the major influence players have in managerial decisions. This includes not just the automatic runner, the batter, and the pitcher in place to start the inning, but also the next hitters to come to the plate any bench players available to be substituted in as a pinch hitter or pinch runner, and pitchers available to be brought in from the bullpen. For example, if Aaron Judge, the 2024 and 2025 American League Most Valuable Player, is the player beginning the inning at bat, a manager would not have him bunt even if it would increase the probability of his team scoring a run. Similarly, if Aroldis Chapman, the 2025 American League Reliever of the Year, is on the mound to begin the inning, a manager may opt to have his player attempt to bunt the automatic runner over to third base to increase the chances of that runner scoring in the inning. Future analysis can seek to factor in these personnel considerations to build a more holistic view of determining winning strategies in extra innings.

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